

Dr. Thyagarajulu Gollapalli

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Research Interests

Numerical Modelling, Machine/Deep Learning, High Performance Computing, GIS, Megathrust Seismicity, Groundwater Modelling, Geophysical Data Acquisition, Processing, and Interpretation

Current Employment

Research Officer/Software Engineer

Australian National University

Canberra, ACT

Jan. 2026 - Present

- Open-source developer for the Underworld codebase, contributing to core features and user-facing interfaces while enhancing modelling capabilities through integration of geological and geophysical processes. Leveraged AI-assisted tools to accelerate development, testing, and debugging, improving code quality, research productivity, and reproducibility.

Work Experience

Assistant Lecturer (Maternity Cover)

Monash University

Melbourne, VIC

Jun. 2025 - Dec. 2025

- EAE3032 (Undergraduate) – Geophysical Interpretations in geoscience, geography and environmental science
- EAE5053 (Postgraduate) – Advanced Spatial Analysis and Modelling

Senior Research Officer/Software Engineer

Monash University and AuScope

Melbourne, VIC

Aug. 2023 - Jun. 2025

- Responsible for high-quality research computing services for the AuScope Simulation, Analysis and Modelling (SAM) node, including core development of the Underworld codebase and user-facing interfaces for the geodynamics community; led testing and debugging, prepared user documentation, and organised training workshops.

Imaging Geophysicist

CGG

Perth, WA

Nov. 2022 - Jun. 2023

- Applied advanced subsurface imaging to resolve complex geology and enhance image quality; designed, executed, and monitored end-to-end seismic processing with rigorous QC throughout; managed schedules and HPC resources to manage deliverables on time while communicating progress and risks to stakeholders.

Education Background

Monash University & Indian Institute of Technology (IIT) Bombay

Doctor of Philosophy, Computational Geodynamics & Geophysics

Melbourne, VIC

Apr. 2018 - Nov. 2022

Thesis: Unravelling tectonic coupling and loading along Sunda margin through 3-D regional numerical modelling

Advisors: Fabio A. Capitanio, Monash University & M. Radhakrishna, IIT Bombay

Indian Institute of Science (IISc)

Bachelor of Science, Major: Mathematics,

Minor: Earth & Environmental Science

Bengaluru, India

Aug. 2011 - May 2015

Thesis: One-dimensional totally asymmetric simple exclusion process

Advisor: Arvind Ayyer, Department of Mathematics

Research Experience

Department of Earth Sciences, IIT Bombay

Junior Research Fellow in Gravity & Magnetism Lab

Mumbai, India

May 2017 - Apr. 2018

Research: Structure and Rheological characteristics of the lithosphere below the Northern Indian Ocean based on the Gravity-Geoid modeling.

Advisor: M. Radhakrishna

Centre for Earth Sciences, IISc

Project Assistant in Computational Geodynamics Lab

Bengaluru, India

Aug. 2015 - Mar. 2017

Research: Investigating Indian ocean geoid low.

Advisor: Attreyee Ghosh

Centre for Earth Sciences, IISc

Undergraduate Research Assistant in Thin section Lab

Bengaluru, India

May 2014 - Jul. 2014

Research: Delineating metamorphism effect in rock minerals along chilled margins.

Advisor: Kusala Rajendran

Centre for Earth Sciences, IISc

Undergraduate Research Assistant in Mass Spectrometry Lab

Bengaluru, India

May 2013 - Jul. 2013

Research: Latitudinal dependency of carbonates from dental enamel.

Advisor: Prosenjit Ghosh

Department of Mathematics, IISc

Undergraduate Summer Intern

Bengaluru, India

May 2012 - Jul. 2012

Research: Group theory reading and discussion sessions.

Advisor: Kaushal Verma

Publications

1. Khalil, H., **Gollapalli, T.**, Capitanio, F.A., Knight, B., & Brune, S., How Far-Field Rotational Motion Drives Rift Evolution, *Earth and Planetary Science Letters*, (Submitted).
2. Capitanio, F.A., **Gollapalli, T.**, Munukutla, R., Graciosa, J.C., Zuhair, M., Beall, A., & Dal Zilio, L., Bridging the gap between subduction dynamics and the long-term strength of megathrusts, *Nature Communications*, DOI: <https://doi.org/10.1038/s41467-025-65824-7>, 2025.
3. Moresi, L., Mansour, J., Giordani, J., Knepley, M., Knight, B., Graciosa, J. C., **Gollapalli, T.**, Lu, N., & Beucher, R., Underworld3: Mathematically Self-Describing Modelling in Python for Desktop, HPC and Cloud, *JOSS*, DOI: 10.21105/joss.07831, 2025.
4. Graciosa, J. C., Capitanio, F. A., Beall, A., Hargreaves, M., **Gollapalli, T.**, Tang, T., & Zuhair, M., Testing Driving Mechanisms of Megathrust Seismicity With Explainable Artificial Intelligence, *JGR Solid Earth*, <https://doi.org/10.1029/2024JB028774>, 2025.
5. Zuhair, M., **Gollapalli, T.**, Capitanio, F.A., Betts, P.G., & Graciosa, J.C., The role of slab step on tectonic loading along subduction zones: inferences on the seismotectonics of the Sunda convergent margin, *Tectonics*, <https://doi.org/10.1029/2022TC007242>, 2022.
6. Ghosh, A., **Gollapalli, T.**, & Steinberger, B., The importance of upper mantle heterogeneity in generating the Indian Ocean geoid low, *Geophysical Research Letters*, <https://doi.org/10.1002/2017GL075392>, 2017.

Conference Posters & Presentations (selected)

1. **European Geosciences Union General Assembly**, Oral Session, 23-27 May 2022, "Megathrust Seismicity Through the Lens of Explainable Artificial Intelligence".
2. **American Geophysical Union Fall Meeting**, Oral Session, 13-17 Dec 2021, "Subduction dynamics and Tectonic coupling along seismically active margin: the relevance of Trench-parallel forces".

3. **American Geophysical Union Fall Meeting**, Poster Session, 9-13 Dec 2019, "Megathrust Seismotectonics of the Southeast Asian Convergent Margin: Insights from Three-Dimensional Spherical Geodynamic Modelling".

Blog Posts

- Unlocking the secrets of great earthquakes with deep learning, 2022 ([link](#))
- Advanced geodynamic models of giant earthquakes, 2019 ([link](#))

Skills

- **Technical:**
Geodynamic Codes: Underworld, CitcomS, Litmod-3D, ASPECT, GPlates.
Advanced: Python, Matplotlib, Cartopy, Scikit, Plotly, Matlab, AWK, Shell script, Generic Mapping Tools (GMT), Paraview, Microsoft Tools, Git.
Intermediate: QGIS, Surfer, C, Docker, PyCharm, PyVista.
Basic: Fortran, Julia, R, PyTorch, Seaborn.
- **Instruments:** Gravimeter (CG-5 Autograv), Magnetometer (GST-19M Proton Precession), Isotope Ratio Mass Spectrometer (IRMS MAT 253), Thin section & Petrology Lab Instruments.
- **Core:** Data Analysis, Data Visualisation, ML, DL, HPC, GIS, Problem Solving, Leadership, Adaptability, Relationship Building.

Scholarships & Awards

- **Postgraduate Publications Award (PPA)** \$5,151 AUD, Apr-Jun 2022, Monash University.
- **AGU Fall Meeting Student Travel Grant** \$1,000 USD, 13-17 Dec 2021.
- **Graduate Research Completion Award (GRCA)** ~\$5,000 AUD, Dec 2021 - Jan 2022, Monash University
- **Monash Graduate Scholarship** \$27,353 AUD/annum, 2018-2021, Monash University.
- **Faculty of Science Dean's International Postgraduate Research Scholarship** \$176,297 AUD, 2018-2022, Monash University.
- **Junior Research Fellow Scholarship**, 2017-2018, Ministry of Human Resource Development (MHRD), India.
- **INSPIRE Scholarship for Higher Education**, 2011-2015, Department of Science and Technology (DST), India.

Teaching Assistant Experience

- **Physics of the Solid Earth, EAE3532, Teaching Assistant**, 2019, 10 students, teaching, grading, reviews.
- **Earth, Atmosphere and Environment 1, EAE1011, Teaching Assistant**, 2020, ~300 students, teaching, grading, reviews.

Outreach & Service

- **Open Day Demonstrator**, Monash, 2022 & IISc, 2016, 2017
- **Protolith Volunteer**, Biennial festival, IIT Bombay, 2017
- **Member of EAE Outreach Team**, Monash University, 2019

Standardised Tests

- Indian Institute of Technology Joint Entrance Examination (IIT-JEE), 2011, ranked in top 1% of Indian applying for undergraduate study in 2011.
- Graduate Aptitude Test in Engineering (GATE), 2016, ranked in top 20% in Geology & Geophysics stream.

Workshops

- Underworld workshop at Curtin University, 2025.
- Underworld workshop at AuScope HQ, 2024.
- Underworld workshop at University of Sydney, 2023.
- The Victorian Universities Symposium on Computational Geodynamics, 3 Nov 2021. Organised by University of Melbourne.
- Graduate Research Conference, 19 Nov 2019. Organised by Monash University.
- Quantitative Seismology and Microseismicity using Geophysical and Geochemical Techniques, 2017. Organised by CEaS, IISc and Shell Technology Centre, Bengaluru.
- Vijyoshi Camp (annual national science camp), 26-27 Nov 2011. Organised by IISc, Bengaluru.

Interests

Marathon Runner, Cycling, Badminton

- *Nike Melbourne Marathon*, 2021, 42.2 km, Time-03:57:26
- *Wings for Life World Run*, 2022, 18 km, Time-01:43:11
- *Great Ocean Road Marathon*, 2024, 44 km, Time-04:03:32